



Fig. 7: Identifying vegetables

identification process. The ESP AI Thinker cam has a super-bright LED, which is switched on for extra light and it helps in easier detection.

The uploading process takes substantial time, sometimes 7 to 8 minutes. Therefore, have patience while uploading the sketch.

Fig. 6 shows the connections required for this vegetables' detection project. It is built around ESP32 cam (MOD1), SSD1306 OLED (MOD2), LM117 voltage regulator (MOD3), and 5V solenoid with associated relays.

The ESP32 cam detects the articles and the output signal at its GPIO13 pin activates the solenoid. For better performance, you may use a TTGO T camera with ESP32-D0WDQ6 8MB SPIRAM OV2640 camera module, which has 3.3cm (1.3-inch) display, Wi-Fi Bluetooth board, and a Fish-eye lens, which has more RAM. Its camera pins are

already defined in the sketch; just change the camera selection and it will be ready.

After uploading the source code, connect the ESP32 cam as shown in Fig. 6 for the circuit diagram, and also the author's prototype in Fig. 1. Power the ESP32 cam and place the vegetables in front of the cam. It will display the results and move the relays to sort the vegetables.

The author's prototype for identifying vegetables is shown in Fig. 7. This is a rudimentary project, which may need further refining to make it actually useful in a store.

EFY note. For programming ESP32, you can use the FTDI USB programmer. Here we have described a project based for vegetable detection based on TTGO T plus camera. You can make another for pens and pencils detection based on ESP32 camera. Both require similar connections and both project files and codes can be downloaded from electronicsforu.com. **EFY**

EFY Note
The source code
of this project
is available
for free download at
www.electronicsforu.com



Somnath Bera is an electronics and IoT enthusiast working as General Manager at NTPC



Don't Be A Loser!

You don't know
how much you
don't know.

Read

electronics
FOR YOU

to keep up-to-date
with the tech world



To subscribe,
visit efy.in or
contact us at: 011-4059660
Email: support@efy.in

OR

Scan
This
Code



Also available at all leading magazine outlets